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Concrete Experience





United States Army Cyber Center of Excellence

Electromagnetic Warfare Site Selection

112-17ABOLC

OVERALL CLASSIFICATION:

UNCLASSIFIED



Learning Step Activities



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- LSA 1 - Identify Key Requirements of an EW Site
- LSA 2 - Describe EMS Considerations for an EW Site
- LSA 3 - Analyze Terrain and Environmental Factors for the EMS
- LSA 4 - Discuss how to Conduct an EMS Survey



Administrative Information



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Safety Requirements	NONE
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Risk Assessment Level	LOW
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Environmental Considerations	None
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Evaluation	None
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References

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- FM 3-12. *Cyberspace and Electromagnetic Warfare Operations*. 17 Sep 2025.
- ATP 3-12.3, C1. *Electromagnetic Warfare Techniques*. 18 Sep 2024.
- ATP 3-12.4. *Electromagnetic Warfare Platoon*. 09 Jan 2023.
- *Electromagnetic Warfare Training Strategy (EWTS) v7*. 01 Mar 2021.



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Identify Key Requirements of an EW Site



EW Site Requirements

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- EW Teams establish positions to conduct ES and EA missions through the EMS
- These positions are called Electromagnetic Warfare (EW) sites
- EW site is a location from which an EW element can conduct EW tasks such as electromagnetic reconnaissance.
- EW Site Requirements:
 - Line of Sight (LOS) to Objective
 - Accessible
 - Tenable
 - Secure
 - Supports Communications





EW Site Requirements - LOS to Objective



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- Observation and surveillance of the objective through the EMS.
- You don't need to visually see the objective.
- You do need to detect and jam SOIs coming within the objective.





EW Site Requirements - LOS to Objective



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- Your ability to detect and transmit signals is determined by
 - Simulations systems (EWPMT) during the planning phase
 - A site Survey during the preparation and execution phase
 - Continuous assessment throughout the operation
- Planning Documents needed
 - Enemy EOB
 - Enemy SITEMP
 - SOI list
 - Joint Restricted Frequency List (JRFL)
 - Targeting/RECON Guidance





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EW Site Requirements - Accessibility



- Your ability to access it
- If you cannot access the site with your equipment, then you will not be able to use it
- If support assets cannot access the site, then it might be too risky to use
- Accessibility is heavily factored by
 - Vehicle type
 - EW equipment type
 - Support needs
 - Security needs





EW Site Requirements - Tenable



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- Your ability to sustain and support it.
- Without adequate support to the site, the EW teams will become combat ineffective when their supplies run out.
- Untenable sites also run the risk of having elements becoming stuck or isolated when supplies run out.
- For a site to be tenable, you must retain the ability to fall back or break contact from the site.
- **DO NOT PUT A TEAM WHERE THEY CAN BE EASILY ISOLATED!**



EW Site Requirements - Tenable



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- Tenability is heavily factored by
 - Food/Water
 - Equipment Power/Fuel
 - Medical
 - Maintenance
 - Duration of the mission
 - Fallback plan





EW Site Requirements - Security



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- Your ability to provide security.
- The average EW team has M4s and a M249, which provides protection from a team of light infantry or a light skin vehicle.
- Security runs concurrently with risk and risks runs concurrently with probability.
- The potential of the enemy attacking your site.
 - Not easily observable or by highly trafficked locations.





EW Site Requirements - Security



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- Security is heavily factored by
 - Enemy scheme of maneuver
 - Observations and fields of fire
 - Friendly
 - Enemy
- Organic EW team's weapon systems
- External friendly support
- Enemy weapon range
 - Direct
 - Indirect





EW Site Requirements - Communications



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- Your ability maintain communications.
- Maintain the ability to communicate timely and accurately.
- Putting a site closer to the enemy might provide more observation but could degrade your ability to report those observations.
- Enable more decisive operations by providing actionable information.

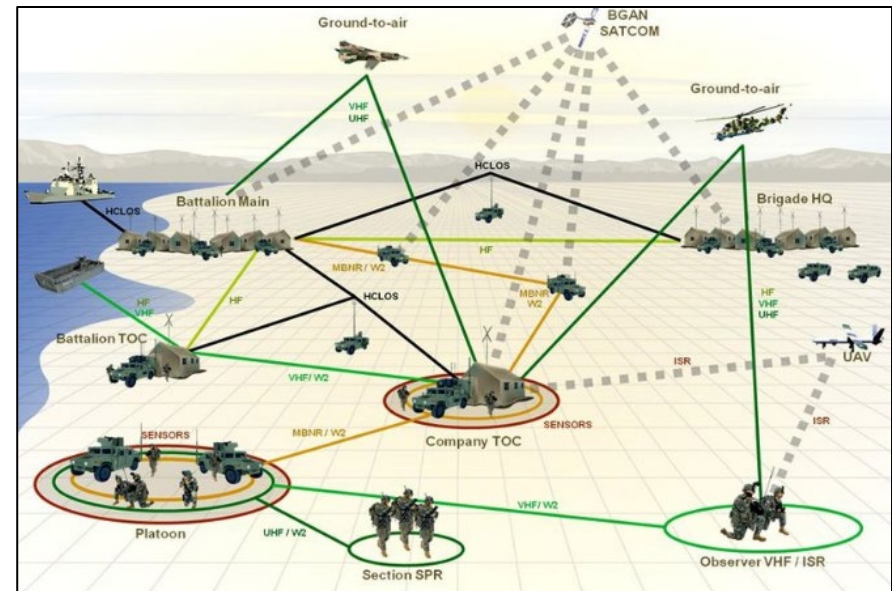


EW Site Requirements - Communications



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- Communication is heavily factored by:
 - Equipment Types, Observations, and fields of fire
 - Friendly
 - Enemy
- PACE
 - BLOS Communications
 - Analog and Data
- Required Reports
- Higher Echelons Comms plan





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Describe EMS Considerations for an EW Site



Key Considerations

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- Requirements vs Considerations
 - The requirements of an EW site serve as a checklist for selection, while considerations allows you to evaluate the requirements.
- Consideration
 - Factors that can directly or indirectly affect your ability to conduct the mission.
 - Can be mission dependent and can change from operation to operation





EW Site Key Considerations



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- Terrain
- Mobility
- Sustainment
- Duration
- External Support
- Equipment Capabilities
- Enemy





Considerations - Terrain

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- Directly affects your ability to access, support and operate from the site.
- Use the acronym OACOK to conduct terrain analysis.
 - Obstacles: What obstacles stand between the EW site and the OBJ?
 - Avenues of Approach: What routes can the EW element take to get to the site?
 - Cover/Concealment: Does the site provide the enemy with visual observation of the EW element?
 - Observations/fields of fire: Is the EW element able to detect SOIs from the location? Is the EW element in range of enemy fires?
 - Key Terrain: What spots must the EW element be able to detect SOIs?



Considerations - Mobility



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- Determines both accessibility and tenability of a site.
- Dismounted
 - Able to access more rugged or isolated sites
 - Limited equipment and supplies
- Mounted
 - Able to take more supplies than dismounted
 - Restricted by terrain
 - Requires extra maintenance and recovery considerations.
- Understand how the supported unit fights vs the EW element's mobility considerations



Considerations - Sustainment



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- Key factor in tenability
- Soldiers and equipment will need a resupply
- “Day of Supply” (DOS)
 - A unit or quantity of supplies adopted as a standard of measurement, used in estimating the average daily expenditure under stated conditions.





Considerations - Sustainment



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- Food/Water
- Equipment Power/Fuel
- Medical
- Maintenance
- Duration of the mission





Considerations - Duration

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- Key factor when conducting an EW mission
- Determines the weight of other factors
 - Ex. The longer you must stay on site, the more important sustainment will become. Meaning tenability will be harder to achieve.
- Duration directly affect considerations:
 - Equipment power
 - Food/Water consumption
 - Fuel consumption
 - Amount of equipment to be carried
 - Security of the site





Considerations - External Support



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- Available friendly support directly affects sustainment, security, and communications.
- An EW PLT should never be completely isolated from external support
- EW PLTs have minimal security capability and no internal sustainment capability.
- The EW PL with the support of the BDE CEMA section, must communicate and coordinate with adjacent and higher echelons to provide support to their EW PLT.
- Understand the different command relationships and who is providing external support to the EW element.



Considerations - Equipment Capabilities



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- Type of EW equipment plays a huge factor into mission execution.
- The EW element must consider EW equipment and communications equipment
- EW Equipment
 - ES/EA capabilities
 - Power output
 - Antenna type
 - Frequency bandwidth
 - Direction finding type





Considerations - Equipment Capabilities



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- Communications Equipment
 - Range between receiving and transmitting nodes
 - Frequency bands used
 - Power output
 - Antenna type

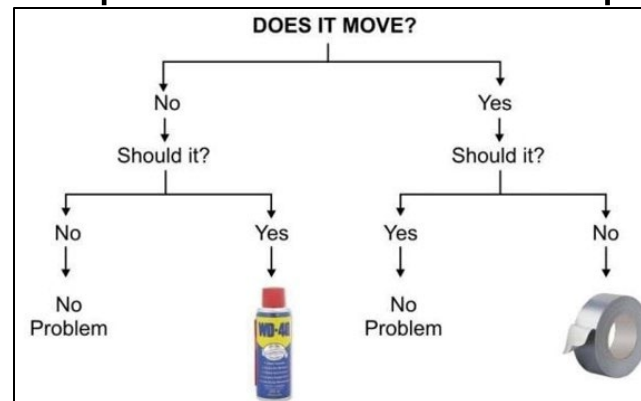




Considerations - Enemy

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- Understand the enemy to learn what the enemy must do and how they will do it.
- Scheme of maneuver (SITEMP)
- Capabilities/limitations of equipment (Enemy EOB)
- Work in conjunction with S-2 prior to and during a mission to provide continuous assessment.
- Be flexible and have multiple courses of action planed.



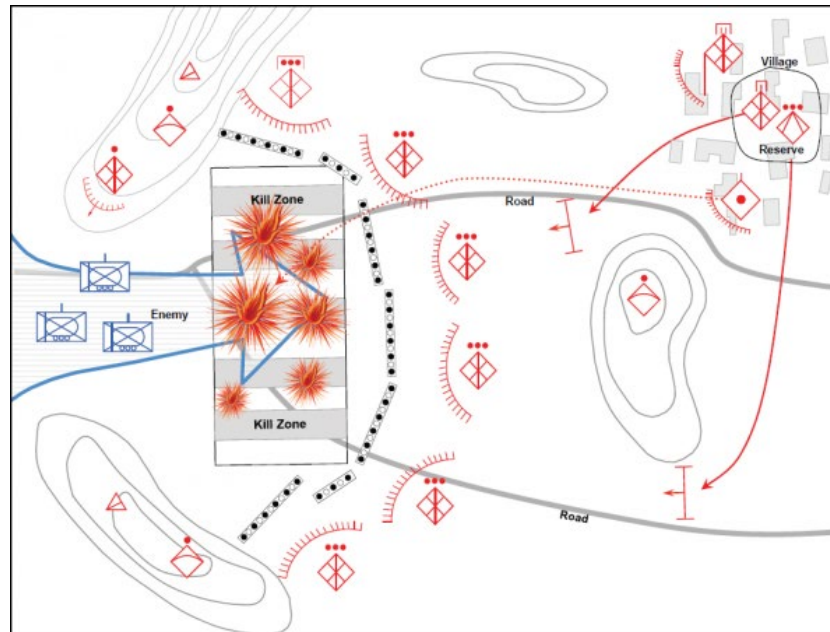


Considerations - Enemy



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- Enemy Scheme of maneuver
- Direct and indirect weapon systems
- EW equipment
- Communications equipment





Key Inputs to Site Selection – Enemy EOB



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- Document that depicts the enemy's capabilities in the cyberspace domain.
- Common Enemy EOBs components:
 - EMS enabled equipment
 - Associated frequencies ranges
 - Number and type of equipment by echelon
- Enemy EOBs can be very detailed or very basic, depending on the information available.
- CEMA personnel work with the S-2/G-2 to refine and improve the enemy EOB

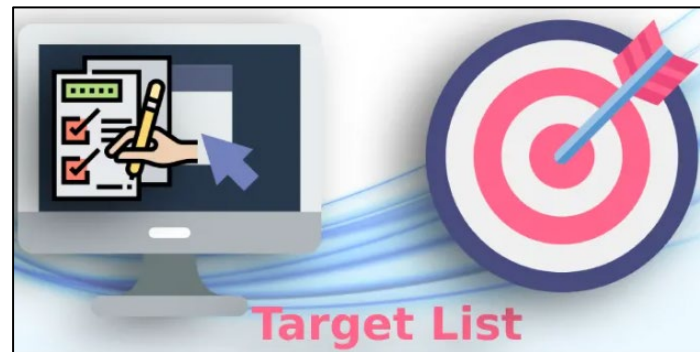


Key Inputs to Site Selection - SOI List



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- Collection of known enemy frequencies with associated equipment.
- SOIs tell you what to look for when conducting EW missions.
- SOI vs Enemy EOB
 - SOI is specific target frequencies list; Enemy EOB is like the enemy SITEMP or Scheme of mauver in the EMS
 - SOI provides the specific avenue of approach for a system; Enemy EOB provides higher level context for that system.





Enemy EOB/SOIs and EW Missions



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- Enemy EOB and SOI list are the most important documents for an EW mission.
- Without the SOI's, EW teams will not be able orientate themselves on their objective within the EMS.
- Without the Enemy EOB, it will be difficult to prioritize information and assess PIRs
- These two documents directly affect your ability to accurately plan for and assess an EW site.
- EW PLT needs frequencies to conduct simulations in planning and to validate the EMS survey



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Enemy EOB Practical Exercise



Enemy EOB Practical Exercise



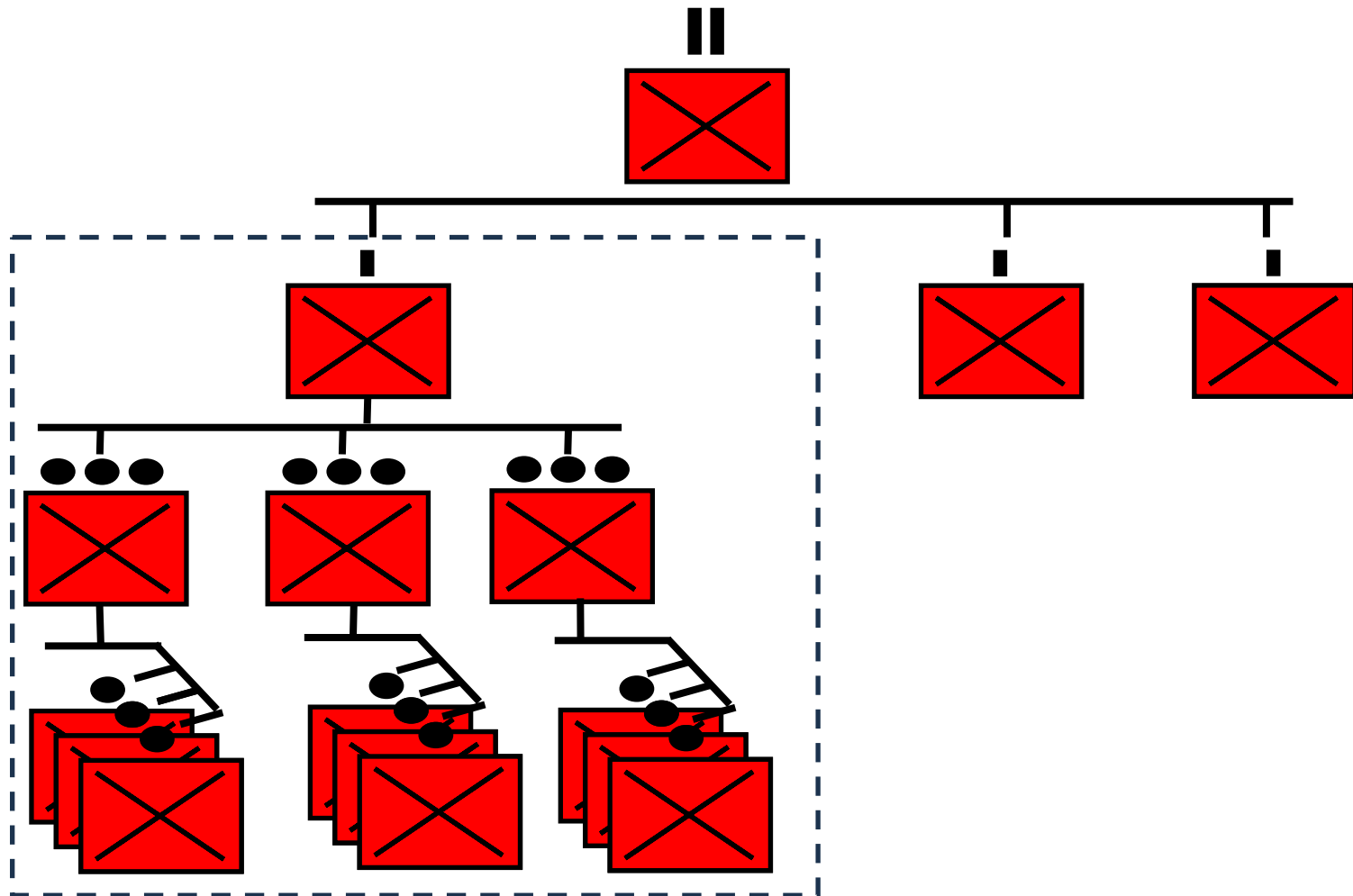
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- Task: Build an Enemy EOB that will help your EW teams identify an enemy infantry company
- Conditions: Individually, given an enemy task organization, communications capabilities, and SOI list
- Standards: In 30 minutes, build an Enemy EOB that clearly depicts communication linkages for the identified enemy company and which communication linkages your teams should target.



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Enemy Task Org Chart





Enemy Communications

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- **Communications Capabilities:**

- VHF COTS (110-200 MHz): Found at all echelons
- VHF Mil Band (30-88 MHz): Found at PLT and above echelons
- UHF COTS SATCOM (400-600 MHz): Found at PLT and CO echelons
- UHF Mil Band SATCOM (500-800 MHz): Found at CO and BN echelons

- **Signals of Interest (SOIs):**

- VHF COTS: 112.5 MHz, 115.5 MHz, 175.0MHz, 185.0 MHz
- VHF Mil Band: 50-55 MHz, 60-65 MHz, 80-85 MHz
- UHF COTS SATCOM: 511.5 MHz, 550.0 MHz, 575.5 MHz
- UHF Mil Band SATCOM: 605.5 MHz, 700.5 MHz, 750.0 MHz



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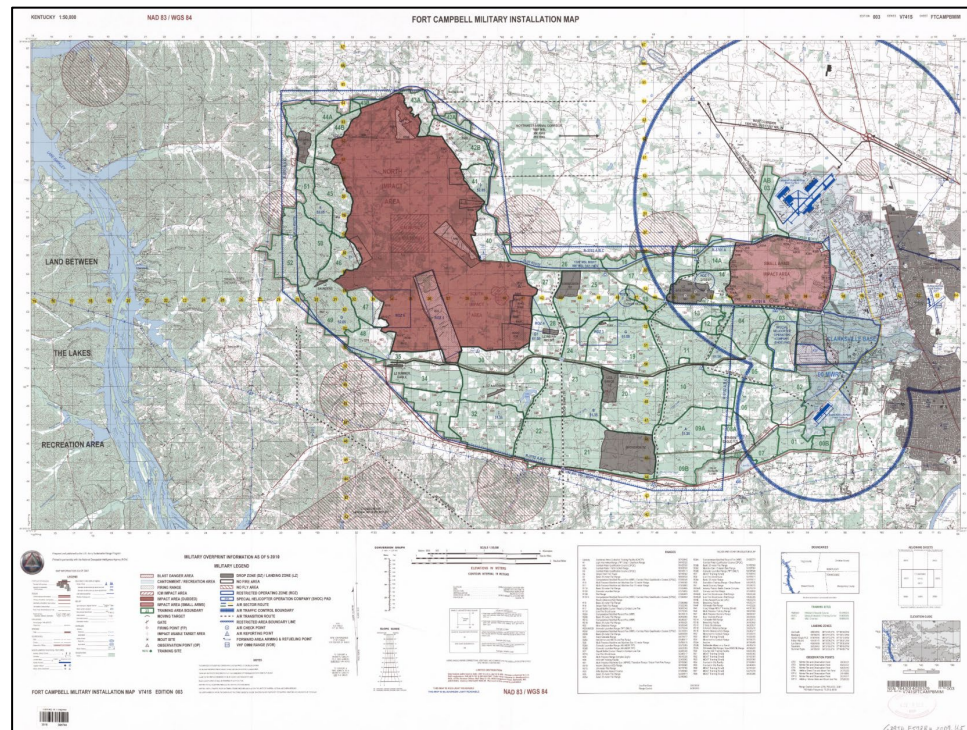
Analyze Terrain and Environmental Factors for the EMS



Map and Simulations

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- Maps
- Most prolific tool for knowledge management at the tactical echelon.
- Map RECON is crucial for selecting an appropriate EW site.

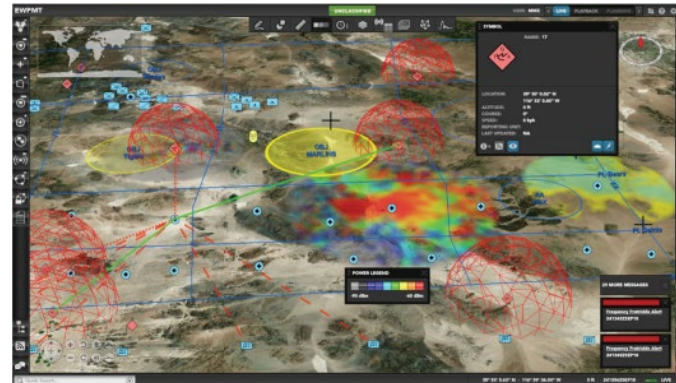




Map and Simulations

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- Simulation
 - Simulations give specific EMS context for terrain analysis
 - Simulation Device:
 - Every EW PLT has one
 - Requires power
 - Duration relies upon battery life
 - Simulation devices in conjunction with a map can be used create an analog product.





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Discuss how to Conduct an EMS Survey



EMS Survey



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- Broad observation of the EMS at a given area.
 - Helps identify activity of signals in designated frequency bands.
- Uses a spectrum analyzer to “Sweep the EMS” and detect active signals.
- Detected signals are checked against operational products to determine friend, neutral, or foe.
- Enemy signals detected during the survey will be documented and/or saved on the device, then briefed to the EW team.
- EMS surveys will help confirm or deny planning assumptions
- An EW site is not established until an EMS survey is conducted.

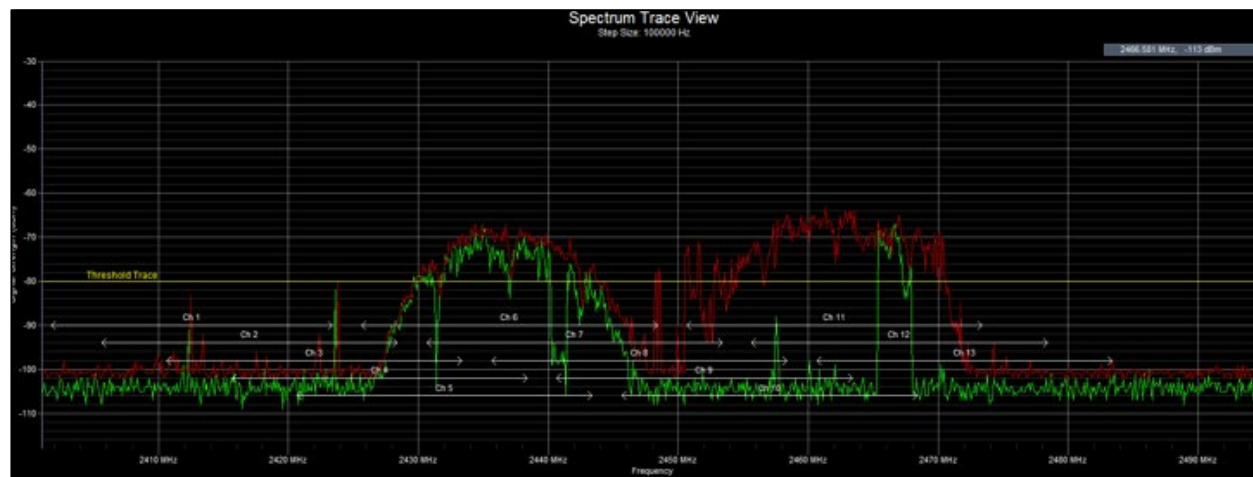


Site Survey Group Discussion



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- Task 1: Take 10 Mins to read through the Conduct an EMS Survey (Digital/Analog) task from the Electromagnetic Warfare Training Strategy (EWTS) v7
- Task 2: Take 10 Mins to discuss in groups of 5, what important tasks you identified and what planning/preparation inputs you would need in order to do those tasks i.e. (products, equipment PCCs and PCIs...)





EWTS - Conduct an EMS Survey (Digital/Analog)



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Section: Individual EW Soldiers (All skill-levels)

Tier: 4 – Individual Level Capabilities

Table: III – Execute

References: ATP 3-12.3, FM 3-12, JP 3-85

Task/Task Number(s): [TBD]

Planning Considerations:

- ☐ Coordinate for a field or foundry training site
- ☐ Ensure location includes environmental and topographic factors similar to the expected OE (i.e. terrain, vegetation, urban features, etc.)
- ☐ Coordinate for support personnel or signal transmitters to be in the targeted area and prepared to generate signals in an obvious manner simulating a live signal environment
- ☐ Coordinate for any additional resources as required (e.g. IEWTPT)
- ☐ Recommended time to complete this task is 3 hours per Soldier

Resource Requirements:

- ☐ Assigned dismounted systems with spectrum analyzer capability
- ☐ Any appropriate safety and personal protective equipment

Conditions: You are an EW Soldier with your assigned dismounted equipment (with spectrum analyzer capability) in a permissive, live-signal, field environment.

Standards: Survey your assigned spectrum band identifying and characterizing each signal currently in your view. Capture signal characteristics of the emissions you find and list them into a single report.

Evaluator Execution: Present Soldier with the task, conditions, and standards. Provide Soldiers a radio band or range to target and an operating environment with multiple live and constant emissions. The Soldier will identify each emission's center frequency, bandwidth, signal-to-noise ratio (SNR), shape, visual characteristics, associated harmonics, and behavior. Keep the report generated by this task for later use in task 4.IV.2.

Evaluation Guidance: Grade the certification according to Performance Steps table. Blacked-out steps/sub-steps provide structure and/or reference for the evaluator. Mark each performance step either GO or NO-GO. The Soldier must complete all performance steps correctly and safely to receive a GO. If any of the Soldiers fail any performance steps identify all errors and instruct the Soldier how to accomplish each performance step correctly.



EWTS - Conduct an EMS Survey (Digital/Analog)



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1. Determine and move to a suitable location to have a good “view” of the EME
2. Note the survey and environment characteristics
 - a. Terrain type/vegetation
 - b. Location/current grid
 - c. Weather
 - d. Temperature
 - e. Season
3. Set the spectrum analyzer sweep to its maximum viewable bandwidth or the span of your assigned band, whichever is narrower
4. Set max-hold to identify any signals that may not transmit constantly



EWTS - Conduct an EMS Survey (Digital/Analog)



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5. Capture and characterize the bands: overall activity, number of signals present, areas of major and minor activity, etc.
6. Narrow the view to a manageable width for surveying the band/range (i.e. ~ 20-50 MHz)

Note: The spectrum view-width chosen in step 6 must be narrow enough to allow for the identification of target signals, yet wide enough to catch all signal harmonics and neighboring emissions

7. Step through the band/range capturing and characterizing each emission found.
 - a. If applicable, adjust sensor waterfall display amplitude as necessary to capture emission behaviors



EWTS - Conduct an EMS Survey (Digital/Analog)



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Note: The view width (e.g. 50 MHz) selected in step 6 must be annotated in the survey and preserved while stepping through the spectrum to maintain consistency, clarity, relativity, and repeatability

- b. Captures should be taken twice:
 - i. One centered on the emission with the view set at your chosen survey width
 - ii. Again centered on the emission with the view set a few kHz wider than the signal's bandwidth



EWTS - Conduct an EMS Survey (Digital/Analog)



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- c. Characterize each emission's dimensions and visually identifiable information:
 - I. Center frequency
 - II. Bandwidth
 - III. Signal-to-noise ratio
 - IV. Receive power level
 - V. Characteristics, span, and dimensions of any visual side-lobes or signal harmonics
 - VI. Description of the signal's basic shape
 - VII. Noticeable behavior
 - VIII. Any additional visually identifying characteristics



EWTS - Conduct an EMS Survey (Digital/Analog)



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8. Log all emission information and spectrum captures in a survey log
9. Repeat steps 7 & 8 for each section of the band, maintaining the survey view width and documenting each emission found
10. Log and save the final report for later development and assessment



Summary



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- Identified Key Requirements of an EW Site
- Described EMS Considerations for an EW Site
- Analyzed Terrain and Environmental Factors for the EMS
- Discussed how to Conduct an EMS Survey



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Questions